# Tugas Morphology

Penulis: Kenny Gabriel Manurung
Dosen: Arta Kusuma Hernanda
NRP : 5024241022
Nama Kelas: Pengolahan Citra dan Video (A)

## Kode yang Digunakan

```
import cv2
import numpy as np


# 1. MEMBACA GAMBAR

img = cv2.imread("rb.png")

if img is None:
    print("Error: gambar rb.png tidak ditemukan!")
    raise SystemExit


# 2. MENGUBAH GAMBAR MENJADI GRAYSCALE

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)


# 3. MENGUBAH GRAYSCALE MENJADI CITRA BINER

_, binary = cv2.threshold(
    gray,
    0,
    255,
    cv2.THRESH_BINARY + cv2.THRESH_OTSU
)


# 4. MEMBUAT KERNEL 5 x 5

kernel = cv2.getStructuringElement(
    cv2.MORPH_RECT,
    (5, 5)
)


# 5. OPERASI EROSI

erosion = cv2.erode(
    binary,
```

```
    kernel,
    iterations=1
)


# 6. OPERASI DILASI

dilation = cv2.dilate(
    binary,
    kernel,
    iterations=1
)


# 7. OPERASI OPENING

opening = cv2.morphologyEx(
    binary,
    cv2.MORPH_OPEN,
    kernel
)


# 8. OPERASI CLOSING

closing = cv2.morphologyEx(
    binary,
    cv2.MORPH_CLOSE,
    kernel
)


# 9. MENAMPILKAN GAMBAR

cv2.imshow("1. Gambar Asli", img)
cv2.imshow("2. Grayscale", gray)
cv2.imshow("3. Citra Biner", binary)
cv2.imshow("4. Erosi", erosion)
cv2.imshow("5. Dilasi", dilation)
cv2.imshow("6. Opening", opening)
cv2.imshow("7. Closing", closing)

print("Tugas 5 - Morphology")
print("Kernel      : 5 x 5")
print("Iterations  : 1")
print("Operasi     : Erosi, Dilasi, Opening, Closing")
print("Tekan tombol apa saja pada jendela gambar untuk keluar.")

cv2.waitKey(0)
cv2.destroyAllWindows()
```